

QUESTION-1:

Title: AUTONOMOUS DELIVERY SCHEDULING OPTIMIZATION (GREEDY + SCHEDULING)

Aim:

To analyze how scheduling algorithms optimize autonomous delivery operations, minimize delivery delays, and maximize resource utilization while satisfying deadlines and operational constraints.

Introduction:

Autonomous delivery services use vehicles or robots to deliver packages within specified deadlines. Efficient scheduling is essential to reduce delivery time, improve customer satisfaction, and maximize vehicle utilization. Greedy scheduling algorithms are commonly used because they provide fast and near-optimal solutions for real-time delivery environments.

Problem Analysis:

The delivery system receives multiple orders, each having:

- Pickup location
- Delivery location
- Delivery deadline
- Package size or weight

The objective is to schedule deliveries so that the maximum number of orders is completed before their deadlines while minimizing travel time and operational cost.

Constraints:

1. Time Windows

Orders must be delivered within specified deadlines.

2. Vehicle Capacity

Each vehicle has limited carrying capacity.

3. Route Distance

Longer routes increase delivery time and fuel consumption.

4. Traffic Conditions

Travel times may vary due to congestion.

5. Fleet Size

Limited number of autonomous vehicles available.

Scheduling Solution:

Greedy Strategy:

- Step 1: Sort delivery requests by earliest deadline.
- Step 2: Assign deliveries to available vehicles.
- Step 3: Select the next delivery with the minimum deadline.
- Step 4: Update vehicle schedule after each assignment.
- Step 5: Repeat until all deliveries are scheduled.

This approach prioritizes urgent deliveries and reduces the chance of missing deadlines.

Real-Time Adaptability:

The system can dynamically adjust schedules when:

- New orders arrive.
- Traffic conditions change.
- Vehicle failures occur.
- Delivery priorities are updated.

Greedy algorithms support rapid rescheduling because decisions are made using current system information.

Computational Complexity:

Sorting Deliveries: $O(n \log n)$

Assignment Operations: $O(n)$

Overall Complexity: $O(n \log n)$

This complexity makes the solution suitable for large-scale logistics systems.

Results and Logistics Efficiency:

Benefits include:

- Reduced delivery delays
- Increased vehicle utilization
- Faster order completion
- Improved customer satisfaction
- Lower operational costs

Conclusion:

Greedy scheduling algorithms provide an efficient and scalable solution for autonomous delivery systems. By prioritizing deliveries based on deadlines and continuously adapting to real-time conditions, logistics efficiency can be significantly improved.